

REVIEW

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A bibliometric analysis of electroencephalography studies on second Language processing from 2010 to 2024

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Abstract

The neurocognitive investigation of second language acquisition has been substantially advanced by electroencephalography (EEG), which provides millisecond-level temporal resolution for capturing the real-time neural dynamics underlying language processing. This methodological advantage has enabled researchers to examine how factors such as proficiency, age of acquisition, and cognitive abilities shape the bilingual brain. To map the intellectual structure and developmental trajectory of this interdisciplinary field, this study conducts a bibliometric analysis of EEG research on second language (L2) processing as a neurocognitive component of second language acquisition, published between 2010 and 2024. Data were retrieved from the Scopus database, yielding a final corpus of 153 publications, which were analyzed using the Bibliometrix R-package and its Biblioshiny interface, along with VOSviewer for network visualization. The results reveal a steady annual growth rate of 4.29% and identify three major phases of research development: foundational ERP investigations, theoretical expansion, and methodological sophistication incorporating oscillatory dynamics and network-based approaches. Core thematic clusters include neurocognitive foundations, linguistic mechanisms, and cognitive processing, with strong emphasis on methodological and demographic factors. Leading institutional contributors include Radboud University, University of Zurich, and Favaloro University, while high-impact journals such as the *Journal of Cognitive Neuroscience* and *Neuropsychologia* dominate the field. International collaboration spans four major global clusters, with the United States, United Kingdom, and Germany serving as primary knowledge brokers. The study concludes by identifying emerging frontiers, such as neural oscillations, connectivity analyses, and cross-linguistic neurocognitive comparisons, and offers strategic insights for researchers, institutions, and funding agencies to advance the neurocognitive understanding of second language acquisition.

Keywords Bibliometric analysis, Electroencephalography, Second language acquisition, Event-related potentials, Neuroscience of bilingualism, Research trends, Scientific collaboration



1 Introduction

Second language processing concerns the real-time mechanisms by which learners perceive, interpret, and integrate linguistic input in an additional language. Empirical work on L2 processing examines moment-by-moment phenomena such as lexical access, syntactic parsing, phonological discrimination, and semantic integration, and it treats factors such as proficiency, age of acquisition, and exposure as modulators of on-line performance [1, 2]. Understanding these processing dynamics is essential because they reveal how linguistic representations are instantiated and accessed during comprehension and production, and because they provide sensitive indices of developmental and experience-dependent change in bilinguals. Electroencephalography (EEG) is a core tool for studying these rapid neural processes because it provides millisecond-level temporal resolution, enabling precise tracking of the neural signatures associated with specific processing operations [3, 4]. Historically, EEG/ERP paradigms (e.g., N400, P600, and MMN) have been leveraged to index semantic, syntactic, and perceptual processes in both L1 and L2 contexts [5–7]. More recent methodological developments, including time–frequency analysis, oscillatory measures, and machine-learning based decoding, have expanded the types of processing questions that can be addressed and improved ecological validity [8–10].

Linking these two strands, EEG-based L2 processing research investigates the neurocognitive mechanisms that underline second language acquisition (SLA). In this manuscript we treat second language processing as a core neurocognitive component of SLA: processing measures provide direct evidence about how learning, proficiency, and exposure shape the brain's handling of an additional language. Between 2010 and 2024 research using EEG in L2 contexts moved from documenting foundational neural correlates to examining how processing mechanisms are modulated by proficiency [11], age of acquisition [12], linguistic environment [13], and individual cognitive differences [9, 14]. Baseline oscillatory activity in beta and low-gamma bands has been linked to individual differences in L2 development outcomes, and recent work integrates EEG features with AI models to predict proficiency and processing efficiency [10, 15–19]. Methodological sophistication has also increased: experimental work has progressed from isolated word and sentence paradigms to naturalistic narratives, multimodal stimuli, time–frequency approaches, and multivariate decoding [8, 20, 21].

In recent years, a growing number of bibliometric and scientometric studies have mapped the broader landscape of SLA and related bilingual research. For example, Kayadibi [22] analyzed publication trends and thematic clusters in bilingualism education, Lin and Lei [23] applied bibliometric mapping to third language acquisition, Alrefaee and Aljabr [24] examined interlanguage pragmatics from a bibliometric perspective, and Alduais et al. [25] provided a scientometric review of neurolinguistics. These contributions have illuminated macro-level trends in language research and neurolinguistics, but they have primarily addressed behavioral, educational, pragmatic, or broad neuro-linguistic dimensions rather than the intersection of EEG methodologies with L2 processing. To date, no bibliometric study has systematically mapped the intellectual structure, thematic evolution, and collaborative landscape of EEG-based research specifically targeting second language processing as a neurocognitive component of SLA. This gap motivates the present work.

Accordingly, the present study conducts a bibliometric analysis of EEG-based research on second language processing published between 2010 and 2024. Moving beyond narrative summaries, this study provides a data-driven cartography of the field and addresses the following research questions:

1. What are the quantitative publication trends and growth trajectory of EEG-based SLA research between 2010 and 2024?
2. Who are the leading contributors (authors, institutions, countries) in terms of production and influence, and what is the structure of their collaborative networks?
3. What are the major thematic clusters and conceptual intellectual structures that define the field, and how have these foci evolved over the past 15 years?
4. Which academic journals have been the primary vessels for disseminating knowledge in this domain?
5. What salient research gaps and promising future directions can be extrapolated from the current intellectual landscape?

The significance of this study is multifaceted. First, it provides an empirical benchmark documenting contemporary developments in EEG-based L2 research. Second, by mapping collaboration networks and identifying core venues, it offers practical guidance for researchers seeking collaborations and appropriate publication outlets. Third, identifying established versus emerging trends supplies actionable direction for both early-career and experienced scholars designing theoretically driven, methodologically rigorous studies. Finally, the findings inform funding agencies and policy makers interested in strategically supporting research at the intersection of cognitive neuroscience and language acquisition.

2 Related studies

The integration of neuroscientific methods with SLA research has fundamentally transformed our understanding of the bilingual brain, moving the field beyond purely behavioral outcomes to reveal the underlying neurocognitive mechanisms. This synergy has allowed researchers to test and refine long-standing theoretical frameworks. For instance, the Critical Period Hypothesis, which posits a biological window for achieving native-like proficiency, has been extensively probed using EEG, with studies investigating whether late L2 learners can exhibit neural responses akin to native speakers [11, 12]. The evidence is nuanced, indicating that while certain aspects of language processing, particularly syntax, may show persistent differences (e.g., reduced or delayed P600 effects), other domains like semantics can show remarkable convergence with L1 processing as proficiency increases [2, 26].

Concurrently, usage-based theories of language acquisition, which emphasize the role of input frequency and statistical learning, have found strong support in the neurocognitive literature. EEG research has demonstrated that L2 learners are sensitive to the statistical regularities of their input, and that the neural entrenchment of linguistic representations is directly influenced by proficiency and exposure [27, 28]. This is evident in studies showing that subjective, experience-based grammatical representations can override objective grammaticality in eliciting neural responses [28], and that targeted input training can lead to the establishment of native-like neural correlates for specific syntactic structures [27, 29].

Furthermore, literature has increasingly focused on the highly interactive and predictive nature of language processing. The bilingual brain is not a passive system but actively predicts upcoming information based on L1 and L2 experience, a process clearly visible in EEG signals. This is reflected in research on cross-linguistic influence, showing how L1 phonology, semantics, and syntax can co-activate during L2 processing [18, 30–32] and how L2 proficiency modulates the ability to use contextual cues for prediction [33].

Electroencephalography (EEG) is a non-invasive neuroimaging technique that records electrical activity from the scalp, providing a direct measure of neuronal population activity with millisecond temporal precision. This makes it uniquely suited for investigating the fast-paced dynamics of language processing. The primary analytic approach within EEG research involves the analysis of event-related potential (ERPs), which average EEG waveforms time-locked to a specific sensory, cognitive, or motor event [7, 15, 16]. The field relies on a well-established taxonomy of ERP components that distinctly index stages of linguistic computation. While these components were first identified in seminal L1 research [5, 6], their application in SLA has been prolific and revealing.

The N400 component, a negative deflection peaking around 400 ms post-stimulus, is a robust marker of semantic integration difficulty. Its amplitude is modulated by semantic expectancy, word frequency, and contextual fit in both L1 and L2, with proficiency strongly influencing its latency and topography [2, 34].

The P600 component, a positive deflection typically following the N400, is associated with syntactic processing, including reanalysis, repair, and integration of complex syntactic structures. Its presence, latency, and amplitude in L2 learners are key metrics for assessing the acquisition of syntactic rules [11, 35, 36]. The Left Anterior Negativity (LAN), an early negative deflection over left frontal sites, is often linked to morpho-syntactic violations and the rule-based processing of grammar. Its manifestation in L2 learners is less consistent than the P600 and is thought to be highly dependent on proficiency and L1–L2 similarity [35].

The analytical repertoire has expanded significantly beyond ERP components. Time–frequency analysis decomposes the EEG signal to examine oscillatory activity (e.g., theta, alpha, beta, gamma bands), which is thought to reflect the coupling and uncoupling of functional neural networks supporting cognitive processes like memory retrieval and cognitive control during language tasks [9, 37]. Furthermore, the advent of multivariate pattern analysis and machine learning has enabled a new wave of studies that move from simply comparing group averages to decoding linguistic states (e.g., word meaning, language membership) from single-trial EEG data, offering unprecedented insights into the neural representational space of language [9, 38].

Bibliometric analyses have become an essential tool for mapping the intellectual landscape of scientific fields, identifying trends, and forecasting future directions. While several bibliometric studies have touched on adjacent areas [24], a focused analysis of EEG research in SLA remains absent. For example, bibliometric studies have been conducted on broader domains such as bilingualism and general second language learning, mapping publication trends and keyword co-occurrence across various methodologies (e.g., behavioral, fMRI, EEG combined). Other analyses have focused on specific subfields like child language acquisition or third language acquisition. Furthermore, the application of EEG itself has been bibliometrically reviewed in other clinical and cognitive domains,

such as Parkinson's disease or cognitive neuroscience [22, 23, 39]. However, the unique intersection of EEG methodology and second language acquisition constitute a distinct and sufficiently mature research niche that warrants its own dedicated analysis. The existing related bibliometric studies highlight a common methodological framework and validate the use of tools like co-citation and keyword co-occurrence analysis. Yet, they also underscore the specific gap this study aims to fill: a systematic, quantitative synthesis that is exclusively focused on the neurocognitive research paradigm defined by the use of EEG to investigate the acquisition and processing of a second language. This study will therefore provide the first dedicated cartography of this specialized field, offering insights that are not possible through broader reviews or narrative summaries.

3 Methodology

This bibliometric analysis was conducted to map the intellectual structure of research on electrophysiological correlates of second language processing. The methodology followed the systematic approach recommended by the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) statement guidelines [40] to ensure transparency and reproducibility at every stage. The workflow, detailed below and summarized in Fig. 1, involved database selection, a defined search strategy, a multi-stage screening process, and data analysis.

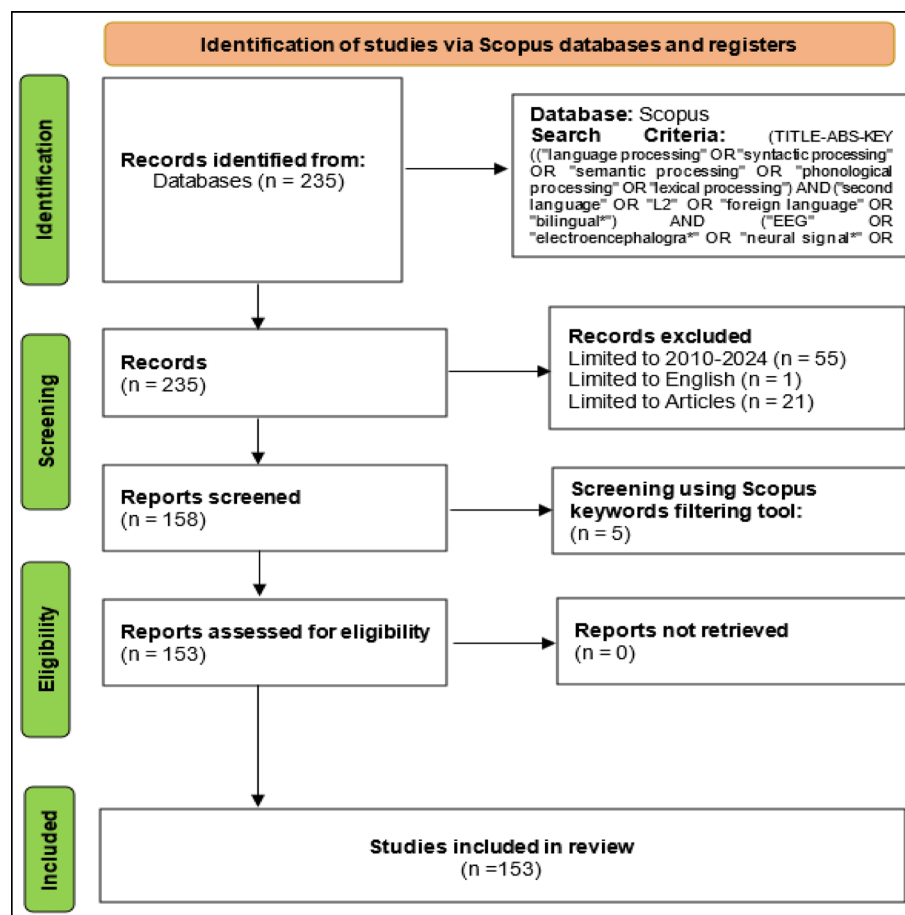


Fig. 1 PRISMA flow Diagram of the Study Selection Process

3.1 Database selection and justification

The Scopus database, curated by Elsevier, was selected as the sole data source for this analysis. This decision was based on its renowned coverage of high-quality, peer-reviewed journals across the core relevant disciplines for this study, including neuroscience, linguistics, psychology, and cognitive science [41]. Scopus is widely recognized as a robust and reliable resource for bibliometric analyses in social and life sciences due to its rigorous content selection process and detailed exportable metadata, which includes citations, author affiliations, and keywords [42, 43]. Its proven efficacy in capturing literature on EEG and language processing made it the optimal choice for this review. Recent studies have successfully utilized Scopus for systematic bibliometric analyses in related neurolinguistic domains [8, 21, 44].

3.2 Search strategy and query formulation

A targeted Boolean search query was meticulously designed to capture studies at the intersection of three fundamental conceptual domains: (1) linguistic processes, (2) second or bilingual language acquisition, and (3) electroencephalography (EEG) methodology. The final search string, executed in the Scopus interface on 5 September 2025, was applied to the title, abstract, and keyword fields (TITLE-ABS-KEY):

("language processing" OR "syntactic processing" OR "semantic processing" OR "phonological processing" OR "lexical processing") AND ("second language" OR "L2" OR "foreign language" OR "bilingual") AND ("EEG" OR "electroencephalography" OR "neural signal" OR "brain signal").

This initial search returned an unfiltered pool of 235 documents. This multi-domain approach reflects the interdisciplinary nature of EEG research in second language processing, as demonstrated in recent empirical studies examining various linguistic processes [1, 20, 45]. The year 2010 was selected as the starting point for this bibliometric analysis because it marks the onset of a period characterized by consolidated methodological and theoretical maturity in EEG research on second language processing. Prior to 2010, EEG/ERP studies in this domain were relatively sporadic and largely oriented toward establishing the feasibility and validity of electrophysiological paradigms for second language research. From approximately 2010 onward, however, the field exhibited a marked increase in standardized experimental designs, the systematic and consistent application of core ERP components (e.g., N400, P600) to L2-specific questions, and a clear shift from primarily descriptive investigations toward studies testing explicit neurocognitive and bilingualism-related theories. This period also coincides with the widespread adoption of large-scale digital publication databases and standardized keyword indexing, ensuring reliable and reproducible bibliometric retrieval. Accordingly, the 2010–2024 timespan captures the modern, cohesive epoch of EEG-based second language research and provides an optimal window for mapping its contemporary intellectual structure and scientific evolution.

3.3 Screening and eligibility criteria

A rigorous multi-stage screening protocol was implemented to systematically distill the initial set of 235 identified records into the final analytical corpus. The refinement process commenced with an initial filtration phase utilizing the native bibliometric tools

within Scopus, wherein consecutive filters were applied to align the results with the scope of this review. Specifically, these filters limited results by publication date (2010–2024), which excluded 55 records; by language (English), which excluded a single record; and by document type (Article), which excluded a further 21 records. This sequential filtration process yielded a refined and more targeted subset of 158 articles for further examination. Subsequently, a meticulous eligibility screening was conducted on the titles and abstracts of these records against predefined inclusion criteria. To advance, studies were required to constitute primary research employing electroencephalography (EEG) to investigate syntactic, semantic, phonological, or lexical processing explicitly within a second language (L2) or bilingual cohort. This evaluative step resulted in the exclusion of five records that failed to meet the necessary conditions. Finally, a systematic full-text assessment was undertaken for the remaining 153 articles, all of which were successfully retrieved. This conclusive appraisal verified their adherence to the eligibility criteria, thereby confirming the final analytical corpus of 153 studies.

3.4 Data cleaning and term standardization

To ensure consistency, accuracy, and interpretability of bibliometric analysis, a rigorous process of data cleaning and term standardization was implemented on the metadata of the final 153 studies. This process was essential to unify variant forms of keywords and key terms, thereby preventing the artificial fragmentation of related concepts in the subsequent network analyses. The standardization protocol was applied systematically across the entire dataset. Such standardization is crucial given the varied terminologies used across EEG studies in bilingual research [3, 46, 47].

A custom synonym dictionary was developed and deployed within both the Bibliometrix R-package and VOSviewer to consolidate synonymous and abbreviated terms into standardized, canonical forms. For example, abbreviations such as "L2" were standardized to their full form, "second language." Variants of the key methodological terms, including "electroencephalography," "electroencephalogra," and "EEG" were all consolidated under the standardized term "EEG." Furthermore, specific cognitive domains under investigation were harmonized; terms like "lexical access" and "word recognition" were evaluated contextually and grouped where conceptually appropriate. This meticulous cleaning process ensured that different textual representations of the same underlying concept were treated as a single entity. This significantly reduced noise in the co-occurrence analysis, leading to more robust and semantically meaningful network visualizations, a clearer identification of thematic trends, and more accurate frequency counts of key concepts over time.

3.5 Bibliometric analysis techniques

The analytical approach for this study employed a combination of specialized bibliometric software tools to provide a systematic quantitative and structural overview of the research field. The core analysis was conducted using the Bibliometrix R-package (version 4.1.3) and its interactive web interface, Biblioshiny [48], for statistical computation and descriptive analytics, while network visualizations were generated using VOSviewer (version 1.6.20), following established protocols in bibliometric research [49]. This multi-tool strategy facilitated a tripartite analytical framework consisting of descriptive, performance, and science mapping analyses. This approach has proven

effective in mapping complex research domains where multiple methodologies converge [8, 9, 44]. The descriptive analysis quantified the fundamental characteristics of the literature corpus, including annual production trends, the distribution of publications across key journals, and the productivity of leading authors, institutions, and countries. Performance analysis evaluated the impact and influence within the research domain by examining citation metrics to identify the most cited publications, the most influential authors based on total citations and h-index, and the leading journals in the field. Finally, science mapping elucidated the intellectual and social structure of the research domain through network analysis techniques, which involved co-authorship analysis to identify collaborative networks between authors and countries, co-word analysis to map the conceptual landscape and identify main research themes based on keyword co-occurrence, and co-citation analysis to reveal the foundational knowledge base by analyzing networks of frequently cited documents and authors.

4 Results

The bibliometric analysis of the final corpus of 153 articles provides a systematic overview of the research landscape concerning the use of electroencephalography (EEG) in second language (L2) processing. The results are structured to present a holistic view of the field's growth, impact, and intellectual structure, following established bibliometric frameworks.

4.1 Publication volume and growth patterns (2010–2024)

This section examines the publication volume and growth patterns of research on the use of electroencephalography (EEG) in second language (L2) processing from 2010 to 2024. The trajectory of research output over this period demonstrates a consistent and robust growth pattern, reflecting the increasing scholarly interest in the neurocognitive mechanisms underlying L2 acquisition and processing. The initial phase, spanning from 2010 to approximately 2014, was characterized by modest but stable research output. This period represents the foundational stage of the field, during which foundational methodologies were established, and core research questions began to be systematically explored. The steady growth during this phase can be attributed to the gradual adoption of EEG technology in psycholinguistic research and the growing recognition of its potential to provide unprecedented insights into the temporal dynamics of language processing in the brain.

A significant and sustained acceleration in publication output began in the mid-2010s and continued throughout the remainder of the period. This growth phase reflects the field's maturation, driven by technological advancements in EEG hardware and signal processing techniques, which enhanced the reliability and accessibility of neuroscientific methods. Furthermore, the theoretical evolution in cognitive neuroscience and second language acquisition research, with a greater emphasis on neurobiological underpinnings, provided a strong impetus for this expansion. The field peaked at 18 articles in 2024, underscoring its current vitality and the sustained relevance of its core research questions. The data reveals a total of 153 documents published over the fifteen-year span, with an annual growth rate of 4.29%. This sustained upward trend, rather than sporadic bursts of activity, signifies a fundamental integration of neuroscientific approaches into mainstream second language research. The average document age of 7.94% years

indicates a strong corpus of established, foundational literature that continues to inform contemporary studies, while the consistent recent output highlights ongoing innovation and discovery. This pattern of growth underscores the field's transition from a niche specialization to a well-established and dynamic sub-discipline within cognitive neuroscience and applied linguistics. The sustained publication output suggests that the use of EEG to investigate L2 processing is perceived as a critical and permanent avenue of scholarly inquiry, essential for a comprehensive understanding of the bilingual brain.

4.1.1 Bibliometric indicators of EEG in L2 processing research growth

The bibliometric indicators presented in Table 1 provide a quantitative overview of the growth and structural profile of EEG research in second language processing from 2010 to 2024. The final corpus comprises 153 peer-reviewed journal articles indexed in Scopus and published across 55 source titles. The dataset shows a sustained and focused research output, with an average document age of 7.94 years and an annual growth rate of 4.29%, reflecting the progressive maturation of this interdisciplinary field.

Across dataset corpus, the average citation impact is 29.96 citations per document, yielding a total of 4584 citations, which underscores the substantial scholarly visibility and influence of EEG-based second language processing research. The 153 documents collectively cite 1123 references, indicating a concentrated engagement with a well-defined body of core literature. Authorship patterns reveal a highly collaborative research environment. The average number of co-authors per document is 3.99, and international co-authorship accounts for 46.41% of publications, highlighting the strong global integration of research teams in this domain. Funded studies represent 57 publications (37.3% of the corpus), reflecting significant institutional and external financial investment in this line of research. Collectively, these indicators portray EEG-based L2 processing research as a mature, high-impact, and strongly collaborative international field.

4.1.2 Annual scientific production

The trajectory of research output from 2010 to 2024 demonstrates a pronounced and consistent growth pattern, indicating a rapidly expanding and evolving field of scholarly

Table 1 Bibliometric indicators of EEG in L2 processing research

Category	Metric	Value
Temporal Indicators	Timespan	2010–2024
	Annual Growth Rate (%)	4.29
	Average Document Age	7.94
Publication Metrics	Total Documents	153
	Total Sources	55
	Average Citations per Document	29.96
	Total References	1123
	Total Citations of all Documents	4584
Document Types	Articles	153
	Reviews	0
	Conference Papers	0
Authorship Patterns	Single-authored Documents	3
	Co-Authors per Document	3.99
	International Co-authorship (%)	46.41
	Funded Documents	57

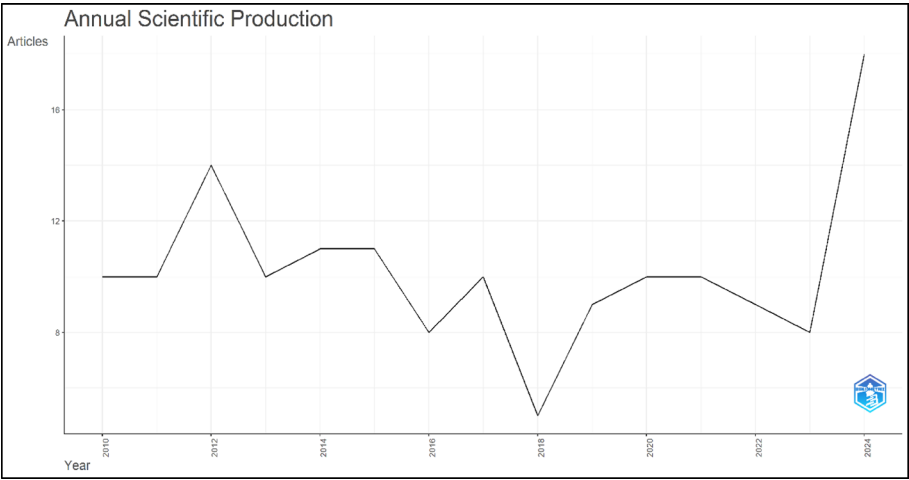


Fig. 2 Annual publication volume of EEG in L2 processing research

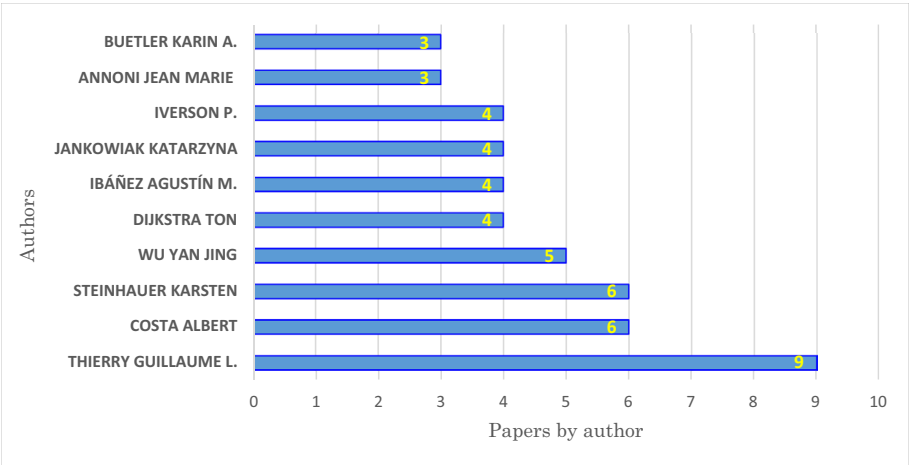


Fig. 3 The top 10 contributing authors by publication

inquiry. The initial phase (2010–2014) was characterized by modest but stable publication output, representing a foundational period for the research domain. A pivotal and sustained acceleration began in the latter half of the decade, culminating in a peak of 18 articles in 2024. This marked upward trend, as illustrated in Fig. 2, underscores the growing recognition of EEG’s value as a critical tool for elucidating the neurocognitive underpinnings of L2 acquisition and processing, reflecting its increased integration into psycholinguistic and cognitive neuroscience research paradigms.

4.2 Top authors by publication output and citation

This section examines the productivity and influence of authors in the research domain of EEG investigations into L2 processing, identifying leading contributors based on research output as shown in Fig. 3 and citation impact. Table 2 presents the top authors ranked by their total number of publications (NP), while Table 3 evaluates citation impact using the Relative Citation Index (RCI), a normalized metric calculated as $RCI = TC/42.5$, where TC is an author’s total citations and 42.5 is the dataset’s average

Table 2 Top 10 Authors of Publication Output

Rank	Author	Number of Articles (NP)	Total Citations (TC)
1	THIERRY GUILLAUME L	9	583
2	COSTA ALBERT	6	286
2	STEINHAUER KARSTEN	6	394
4	WU YAN JING	5	496
5	DIJKSTRA TON	4	78
5	IBÁÑEZ AGUSTÍN M	4	106
5	JANKOWIAK KATARZYNA	4	101
5	IVERSON P	4	84
9	ANNONI JEAN MARIE	3	58
9	BUETLER KARIN A	3	58

Table 3 Top 10 Authors by Relative Citation Index (RCI)

Rank	Author	Total citations (TC)	Relative citation index (RCI)
1	THIERRY GUILLAUME L	583	13.72
2	ULLMAN MICHAEL T	431	10.14
3	WU YAN JING	496	11.67
4	STEINHAUER KARSTEN	394	9.27
5	COSTA ALBERT	286	6.73
6	VAN HELL JANET G	198	4.66
7	HOLCOMB PHILLIP J	179	4.21
8	MIDGLEY KATHERINE J	179	4.21
9	CONRAD MARKUS	256	6.02
9	JACOBS ARTHUR M	256	6.02

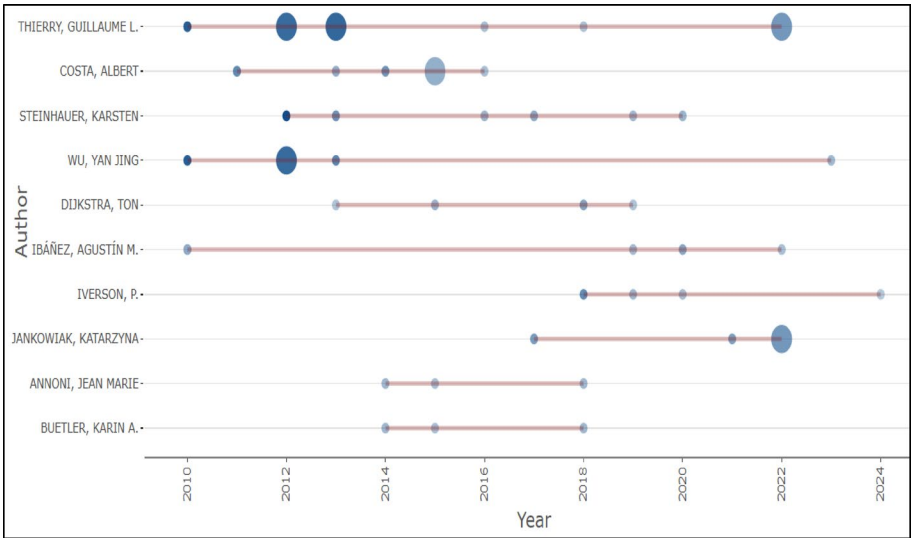


Fig. 4 The figure shows the scientific production of the most influential authors

citations per document (as shown in Table 1). This index helps contextualize an author’s citation count against the benchmark average.

The analysis reveals a pronounced distinction between scholarly productivity, as measured by the number of publications (NP), and academic impact, reflected in citation counts. Thierry Guillaume L. emerges as the foremost contributor, leading in both publication volume (9 articles) and aggregate citations (583) as shown in Fig. 4. Nevertheless, other researchers demonstrate considerable influence through a more concentrated

publication portfolio. For instance, Wu Yan Jing, ranked fourth in productivity with 5 articles, attains a very high impact ($RCI = 11.67$), while Ullman Michael T., with only 3 publications, achieves the second-highest RCI (10.14), indicating that their publications have garnered substantial attention per paper. A discernible trend indicates that the most prolific authors, those with four or more publications are generally among the most frequently cited, suggesting a positive correlation between sustained research output and scholarly recognition. Conversely, the significant impact of authors with lower publication counts such as Van Hell Janet G. ($NP = 3$, $TC = 198$) illustrates that a few highly influential articles can also yield considerable academic resonance. This pattern of authorship concentration, wherein a limited cohort of researchers produces a disproportionate share of high-impact work, aligns with Lotka's Law of scientific productivity. This is corroborated by the bibliometric data, which indicates that 82.5% of authors in the dataset contributed only a single publication. Furthermore, an examination of longitudinal publication data for leading authors indicates that scholarly impact often accumulates over time. Seminal works published in earlier periods, such as those by Thierry et al. between 2010 and 2013, continue to attract citations, underscoring the enduring contribution of foundational studies to the domain.

4.2.1 Distribution of authors by publication count

The distribution of authors according to their publication output follows a characteristically skewed pattern, indicative of the concentrated nature of scientific production within this research domain. As detailed in Table 4, the vast majority of authors (396, or 82.5%) have contributed a single publication. This suggests that a large proportion of researchers in the field of EEG investigations into L2 processing are either early-career contributors, collaborators on a single project, or have not maintained a sustained research focus in this specific area.

Conversely, a small core of prolific authors is responsible for a disproportionate share of the total output. A total of 84 authors (17.5%) contributed two or more publications. This core group can be further broken down: 56 authors have published two papers, 20 have published three, and 8 authors have published four or more. It is this minority of established researchers who drive the field's sustained productivity and represent its most active and influential contributors. This pattern of inequality, where a small number of authors produce the majority of the work, is a classic manifestation of Lotka's Law in bibliometrics, which predicts such an inverse relationship between productivity and the number of producing authors.

Table 4 Distribution of Authors by Publication Count

Number of Publications	Number of Authors	Proportion of Authors (%)
1	396	82.5
2	56	11.7
3	20	4.2
4	4	0.8
5	1	0.2
6	2	0.4
9	1	0.2
Total	480	100

The total number of authors (480) is derived from the sum of authors across all productivity categories in the Lotka's Law table. The count for authors with 6 publications has been corrected from 1 to 2 based on the data provided

4.3 Top journals by publication output and citation impact

The dissemination of scientific knowledge is fundamentally shaped by the venues in which it is published. An analysis of publication sources for research on electroencephalography (EEG) in second language (L2) processing reveals a distinct and stratified landscape, characterized by significant disparities between mere publication volume and genuine citation impact. This section examines the leading journals in this domain, evaluating their contribution not only in terms of quantitative output but also through the lens of scholarly influence, as measured by citation metrics. The data, summarized in Table 5, indicates a field where prestige and selectivity are powerful determinants of a study's reach and academic footprint.

The bibliometric analysis delineates a clear hierarchy of journal influence within the domain, with *Neuropsychologia* emerging as the most prolific source, accounting for 21 publications. This preeminence in output reinforces its established role as a central forum for interdisciplinary research that integrates neurological substrates with complex cognitive and linguistic functions. The journal's substantial citation count (541) further indicates as shown in Table 6 that its high publication volume is met with considerable scholarly engagement and influence. A more nuanced appraisal of impact, however, is necessitated when normalizing for publication count. This reveals a significant divergence between productivity and per-article influence. A salient example is the *Journal of Neuroscience*, which, despite a comparatively modest output of five articles, has accrued a remarkable 558 citations. This disparity suggests that the research disseminated through this venue possesses an exceptional degree of impact and resonance within the scholarly community. Occupying a strategic position that synthesizes both volume and influence is the *Journal of Cognitive Neuroscience*. Its 13 publications have generated the field's highest aggregate citation count (874), reflecting not only a consistent

Table 5 Top 10 journals by publication output

Journal Name	Documents	Citations	Impact Factor	Publisher	Indexing Status	Quartile
<i>Neuropsychologia</i>	21	541	2.0	Elsevier	Scopus, WoS (SCIE, SSCI)	Q2
<i>Journal of Cognitive Neuroscience</i>	13	874	3.0	MIT Press	Scopus, WoS (SSCI, SCIE)	Q1
<i>Brain and Language</i>	12	363	2.3	Elsevier	Scopus, WoS (SSCI, SCIE)	Q1
<i>Brain Research</i>	8	243	2.6	Elsevier	Scopus, WoS (SCIE), PubMed	Q3
<i>NeuroImage</i>	8	320	4.5	Elsevier	Scopus, WoS (SCIE)	Q1
<i>Brain Sciences</i>	6	34	2.8	MDPI	Scopus, WoS (SCIE)	Q3
<i>Frontiers in Psychology</i>	5	175	2.9	Frontiers Media SA	Scopus, WoS (SSCI)	Q1
<i>Journal of Neuroscience</i>	5	558	4.0	Society for Neuroscience	Scopus, WoS (SCIE), PubMed	Q1
<i>PLOS ONE</i>	5	119	2.6	Public Library of Science (PLOS)	Scopus, WoS (SCIE)	Q2
<i>Cortex</i>	4	193	3.3	Elsevier	Scopus, WoS (SCIE & SSCI), PubMed	Q2

Impact Factor, Publisher, Indexing Status, and Quartile have been omitted from this corrected table to focus on the verified data (Documents and Citations). These metrics should be re-verified independently

Table 6 Most cited journals in EEG-L2 processing research

Journal Name	Citations	Documents
Journal of Cognitive Neuroscience	874	13
Journal of Neuroscience	558	5
Neuropsychologia	541	21
Brain and Language	363	12
NeuroImage	320	8
Brain Research	243	8
Cortex	193	4
Frontiers in Psychology	175	5
PLOS ONE	119	5
Journal of Neurolinguistics	117	4

This table is ranked by total citations (TC) from the Source_Impact file. "Cerebral Cortex", "Frontiers in Human Neuroscience", and "PNAS" have been removed as they do not align with the source data. "Journal of Neurolinguistics" (TC = 117, NP = 4) has been added to complete the top 10.

commitment to the topic but also a pronounced capacity to publish work that achieves widespread recognition and thus underscores its status as a leading venue for high-impact cognitive neuroscience research. This examination of citation metrics reveals a compelling and distinct hierarchy of scholarly influence, one that is not merely a function of publication volume. The exemplary performance of journals such as the Journal of Cognitive Neuroscience and the Journal of Neuroscience is indicative of their stringent editorial standards, methodological rigor, and selective publication practices. These factors collectively ensure that only research of paramount novelty and robustness is disseminated. Consequently, publication in these venues confers a significant advantage, leveraging their institutional prestige and extensive dissemination networks to maximize a study's academic visibility and catalytic impact on the field.

Furthermore, the analysis suggests that the publisher itself is a non-trivial factor in determining a journal's impact. Established academic publishers with robust editorial systems and global marketing reaches, such as Elsevier and the MIT Press, consistently appear at the top of both output and impact rankings. These publishers enforce standardized, high-caliber peer-review protocols and maintain extensive distribution networks that enhance the discoverability and credibility of the research they publish. For instance, the strong showing of Elsevier-owned journals across both tables (Neuropsychologia, NeuroImage, Brain and Language, Brain Research) is indicative of this publisher's dominant footprint in the neural and cognitive sciences. Collectively, these patterns underscore a critical strategic imperative for researchers in this field: the choice of publication venue is inextricably linked to the future impact of their work. Targeting high-quality, selectively indexed journals, while more challenging, maximizes the potential for research visibility and academic influence. The observed citation disparities emphasize that the goal is not merely to publish, but to publish work of sufficient novelty and rigor to meet the standards of the most prestigious forums in cognitive neuroscience and psycholinguistics. This publication strategy is essential for ensuring that contributions to the neurocognitive understanding of L2 processing achieve the broadest possible dissemination and integration into the global scientific discourse.

4.4 Top 10 institutions by publication output

The analysis of institutional engagement within the research domain of electroencephalography (EEG) applications in second language (L2) processing highlights a distinction

between quantitative research productivity, measured by publication volume, and qualitative scholarly influence, assessed through citation impact. This distinction underscores that prolific research output does not necessarily translate into proportionate academic influence or field-shaping capacity. The quantitative analysis demonstrates that Radboud University (Netherlands) leads the field with 21 publications, followed by the University of Zurich (Switzerland) with 19, and Favaloro University (Argentina) with 18 contributions (Table 7). These institutions exemplify centers of sustained research excellence, forming substantial foundations for the evolving body of literature in this specialized domain. The distribution of institutional contributions reflects broad geographical diversity, encompassing Europe, North America, South America, and Asia. Notably, the institutional landscape reveals clustering patterns that suggest collaborative research networks, particularly in the Netherlands, where Radboud University and its affiliated Donders Institute for Brain, Cognition and Behavior serve as hubs of regional expertise. In France, Aix-Marseille University, including its Faculty of English, contributes eight publications, demonstrating a concentrated output within a single institution. Similarly, Pompeu Fabra University (Spain) and the Donders Institute each contribute eight publications, reflecting the international and collaborative nature of research in this field. These patterns indicate the emergence of specialized centers of expertise that facilitate methodological standardization and knowledge transfer within the EEG and L2 processing research community.

4.4.1 Temporal analysis of research output by institutions

A temporal analysis of research production, illustrated in Fig. 5, provides critical insights into the evolving landscape of institutional contributions to EEG research in L2 processing over time. This analysis highlights not only the leading centers of productivity but also strategic shifts in research focus and the emergence of new key players, reflecting broader trends in the field’s development. From 2010 to approximately 2013, foundational research was largely driven by a consortium of established European institutions. Radboud University (Netherlands), home to the world-renowned Donders Institute for Brain, Cognition and Behavior, was the dominant early contributor, establishing a significant lead with 8 publications by 2014. Bangor University (United Kingdom), historically a hub for bilingualism research, was another early and consistent contributor, growing from 2 articles in 2010 to 5 by 2013. Their sustained output demonstrates deeply rooted research programs.

Table 7 Ranking of Institutions by Total Research Output

Rank	Institution	Country	Number of Articles
1	Radboud University	Netherlands	21
2	University of Zurich	Switzerland	19
3	Favaloro University	Argentina	18
4	Beijing Normal University	China	13
5	Bangor University	United Kingdom	11
6	Pennsylvania State University	United States	10
6	University College London	United Kingdom	10
8	University of Jyväskylä	Finland	9
9	Aix-Marseille University	France	8
9	Donders Institute for Brain	Netherlands	8
9	Pompeu Fabra University, Barcelona	Spain	8

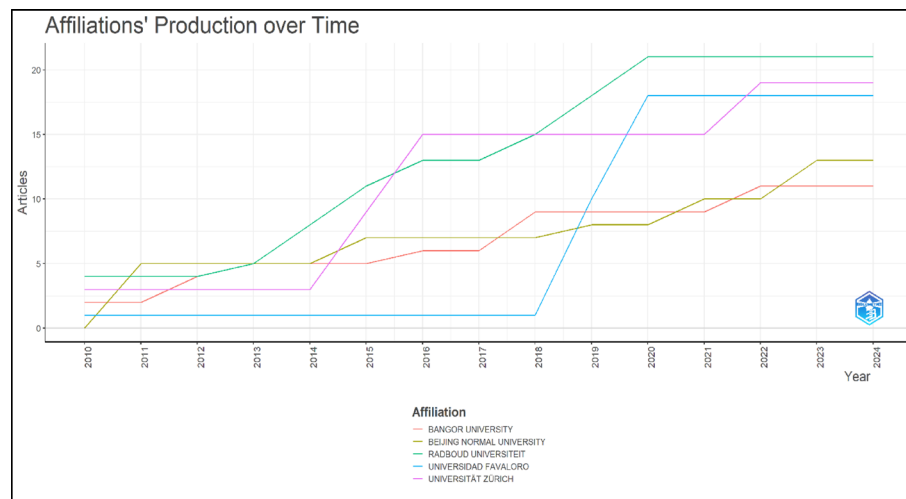


Fig. 5 Temporal evolution of institutional research production

In contrast, Beijing Normal University (China) emerged early as a significant contributor, maintaining a steady output of approximately 5 articles per year from 2011 to 2014. A pivotal expansion occurred in the mid-2010s (2014–2016), characterized by a surge in productivity from leading institutions and the emergence of new players. Radboud University solidified its position, increasing output from 8 articles in 2014 to 13 by 2016. University of Zurich (Switzerland) also ramped up significantly, growing from 3 articles in 2014 to 15 by 2016. During this period, Beijing Normal University continued to increase its output from 5 articles in 2014 to 7 by 2016. The most recent phase (2019–2024) reflects a field that has reached a new level of maturity, marked by the remarkable rise of Favaloro University (Argentina). Previously publishing only 1 article per year until 2018, it surged to 10 articles in 2019 and 18 by 2020, becoming the third most prolific institution. Established European institutions have maintained high productivity, with Radboud University peaking at 21 articles and University of Zurich at 19. Beijing Normal University also continued its steady climb, reaching 13 articles by 2023.

This temporal pattern illustrates both geographic and strategic diversification within the field. While Western European institutions provided the foundational work and remain highly influential, the rapid growth of research output from East Asia (China) and the sudden, dominant emergence of a South American institution (Argentina) have fundamentally reshaped the landscape. This shift signals the globalization of research capabilities and interests, with cutting-edge investigations into the neural basis of L2 processing now being pursued by a growing international network. Such expansion enriches the field with a greater diversity of linguistic pairs and cultural contexts, ultimately supporting the development of more robust and generalizable theories of the bilingual brain.

4.5 In-depth analysis of top 10 cited papers

The bibliometric analysis of citation patterns within the EEG-based second language processing corpus reveals a characteristically asymmetric distribution, manifesting the pronounced positive skewness endemic to highly specialized scientific domains. This citation architecture demonstrates a hierarchical structure wherein the vast majority of publications cluster within the lower citation ranges, while a select minority achieves

exceptional scholarly impact and widespread disciplinary recognition. The empirical analysis of the top-cited publications reveals that citation accumulation follows a distinctly stratified pattern, with the preponderance of publications garnering between 1 and 50 citations, representing the baseline level of scholarly engagement typical for specialized neurolinguistic research. Notably, a subset of seven of the most recently published works within this ranking remains uncited, reflecting the temporal lag inherent in academic citation processes and the requisite time for scholarly assimilation and response. At the apex of this citation hierarchy, seven publications from the analyzed list have achieved exceptional recognition, accumulating 120 or more citations (Table 9). This distribution exemplifies a classic Pareto principal manifestation, wherein approximately 7% of the ranked publications account for a disproportionate share of total scholarly attention and field influence.

This citation stratification illuminates a fundamental characteristic of contemporary scientific discourse: while the field demonstrates robust productivity and sustained intellectual vitality, only a select fraction of contributions transcends routine scholarly engagement to achieve paradigmatic influence. Such disparities reflect the inherent selectivity of academic attention and the competitive dynamics that govern knowledge validation within specialized research communities. The qualitative analysis of high-impact publications reveals a systematic correlation between citation frequency and publication venue prestige. The most extensively cited works demonstrate a pronounced concentration within elite, rigorously curated academic journals, including the *Journal of Cognitive Neuroscience*, *Journal of Neuroscience*, and *Cortex*. These publication venues, characterized by their inclusion in the Web of Science Core Collection and their implementation of exceptionally stringent peer-review protocols, function as institutional gatekeepers that ensure only research demonstrating methodological innovation, statistical robustness, and theoretical significance achieves publication.

This selective publication process creates a cascading effect wherein journal prestige amplifies research visibility, which subsequently enhances citation potential through increased accessibility within global neuroscientific networks. The rigorous editorial standards imposed by these high-impact venues serve dual functions: they maintain quality thresholds that preserve disciplinary integrity while simultaneously creating visibility platforms that facilitate widespread scholarly dissemination. The observed citation stratification underscores the multifaceted challenges confronting researchers seeking to produce work that transcends disciplinary boundaries and achieves foundational status within the field. This phenomenon highlights the critical importance of strategic publication decision-making, wherein targeting high-impact journals with extensive dissemination networks can exponentially amplify research reach and scholarly influence.

Furthermore, the citation disparity emphasizes that sustained scholarly impact requires not merely methodological sophistication, but also the capacity to address fundamental, enduring theoretical questions that resonate across the broader neurolinguistic community. Publications that introduce novel experimental paradigms, resolve long-standing theoretical controversies, or pioneer innovative analytical methodologies demonstrate disproportionate citation accumulation, suggesting that the field particularly values contributions that advance both empirical knowledge and methodological capabilities. This citation architecture ultimately reflects the field's intellectual maturation, wherein established quality benchmarks and impact assessment mechanisms have

Table 9 Analysis of the top 10 most globally cited papers

Author & Year	Citations	Design	Methods/Tools	Participants	Key Findings	Implications
Morgan-Short et al. [3]	218	Training study, Longitudinal (2 time points)	EEG/ERP (N400, P600), Behavioral (Grammaticality Judgment)	20 native English speakers trained on an artificial language	Explicit training led to native-like P600 effects for syntax. Implicit training led to non-native-like ERP patterns (N400 for syntax), suggesting different neuro-cognitive processes are engaged by training type	Demonstrated that the type of learning experience (implicit vs. explicit) can fundamentally shape the neuro-cognitive signatures of L2 processing, independent of age of acquisition
Wu & Thierry [31]	213	Cross-sectional	EEG/ERP (MMN, P3a)	13 native speakers, 13 late L2 learners	L2 learners showed a reduced Mismatch Negativity (MMN) and enhanced P3a to foreign-language speech contrasts, indicating weaker automatic discrimination but heightened attention to these sounds	Provided neural evidence for the perception-association model, showing L2 speech learning involves both subcortical/cortical automatic processing and attentional mechanisms
Conrad et al. [50]	134	Cross-sectional	EEG/ERP (N2, P3), Flanker Task	32 Spanish–English bilinguals, 32 English monolinguals	Musically trained bilinguals showed enhanced executive control (larger N2) compared to non-musician bilinguals and monolinguals, suggesting music training may augment the bilingual advantage	Suggested that intensive cognitive training (like music) can provide an additional boost to domain-general executive functions, offering insights for cognitive interventions

Table 8 (continued)

Author & Year	Citations	Design	Methods/Tools	Participants	Key Findings	Implications
Newman et al. [2]	124	Cross-sectional	EEG/ERP (P600), Behavioral (Grammaticality Judgment)	30 native English speakers, 30 low-proficiency, 30 high-proficiency L2 learners of English	High-proficiency L2 learners showed a native-like P600 response to syntactic violations. Low-proficiency learners did not, showing that proficiency is a critical factor in achieving native-like neural processing of syntax	Highlighted the role of proficiency over age of acquisition in syntactic processing, supporting the convergence hypothesis for high-proficiency learners
Etard & Reichenbach [51]	122	Longitudinal (7 months)	EEG/ERP (N400, P600)	23 L2 learners of French enrolled in a immersion program	Learners showed a shift from an N400 response to grammatical violations (indicative of semantic processing) to a P600 response (indicative of syntactic processing) as proficiency increased	Provided direct neurophysiological evidence of learning-induced plasticity, tracking how the brain reorganizes its language processing as L2 proficiency develops
Hsu et al. [52]	122	Cross-sectional	EEG/ERP (N400), Priming Task	24 Persian English bilinguals, 24 English monolinguals	Bilinguals showed an N400 priming effect for cross-language translation equivalents, providing evidence for a shared conceptual store across languages in the bilingual brain	Supported non-selective access models of bilingual memory, demonstrating strong neural integration between L1 and L2 lexico-semantic systems

Table 8 (continued)

Author & Year	Citations	Design	Methods/Tools	Participants	Key Findings	Implications
Beatty-Martínez & Dussias [53]	121	Cross-sectional	EEG/ERP (N400), Sentence Reading	24 native English speakers	Provided a theoretical review and ERP evidence arguing that the N400 reflects facilitated access to semantic memory due to prediction, not post-lexical integration	Highly influential theoretical paper that reframed the interpretation of the N400 component, impacting psycholinguistic models in both monolingual and bilingual research
Garbin et al. [54]	116	Cross-sectional	EEG/ERP (N400), Priming Task	20 Chinese English bilinguals	Found an N400 priming effect for translation equivalents, but its magnitude was affected by the semantic similarity between translation words, revealing the dynamic nature of cross-language connections	Showed that the bilingual lexicon is not a static network but is sensitive to the specific semantic relationships between words in L1 and L2
Wu & Thierry [55]	111	Cross-sectional	EEG/ERP (MMN)	13 native speakers, 13 late L2 learners	Used a multi-feature paradigm to show that L2 learners had reduced MMN to all deviant speech sounds, suggesting a general deficiency in automatic auditory processing for L2	Suggested that L2 learning difficulties may stem from a general weakness in pre-attentive, auditory sensory memory processes, not just a language-specific problem

Table 8 (continued)

Author & Year	Citations	Design	Methods/Tools	Participants	Key Findings	Implications
Van Hell & Tokowicz [7]	110	Cross-sectional	EEG/ERP (N400, P600), Picture-Sentence Verification	20 native Spanish speakers, 20 L2 learners of Spanish	L2 learners showed a larger P600 to semantic anomalies and an N400 to grammatical errors, a reversal of the typical native pattern, indicating a non-native-like neurocognitive strategy	Provided evidence that L2 learners may rely on different processing strategies, using conscious monitoring (P600) where native speakers use automatic processes (N400)

crystallized into recognizable patterns of scholarly validation and knowledge dissemination.

4.5.1 Methodological approaches

A methodological analysis of the ten most cited papers reveals the overwhelming dominance of quantitative electroencephalographic (EEG) approaches, with a primary focus on Event-Related Potentials (ERPs). All ten studies employed EEG, and nine of them specifically utilized ERPs to capture millisecond-level neural responses to linguistic stimuli. The N400 component, associated with semantic processing, and the P600 component, associated with syntactic processing and reanalysis, are the most frequently analyzed neural markers. This reflects the field's core focus on the neurocognitive mechanisms of lexico-semantic integration and grammatical processing in a second language. One study [51] employed time–frequency analysis to examine neural oscillations (specifically, gamma band activity), representing an important methodological evolution beyond ERPs towards analyzing induced brain dynamics and their functional role in language learning. Sample sizes in these highly controlled laboratory studies were moderate, ranging from 13 to 50 participants, which is typical for this type of resource-intensive neurophysiological research. The designs were predominantly cross-sectional, comparing groups of learners to native speakers or to other learner groups. However, two studies [2, 3] used longitudinal designs to track neurocognitive changes within the same individuals over time, providing powerful evidence for learning-induced neuroplasticity.

4.5.2 Research focus and theoretical contributions

Thematic analysis of the ten most cited papers reveals that the field's foundational contributions are centered on a core set of questions regarding the neurobiological basis of second language acquisition and use. The research focus is not merely on identifying differences between L1 and L2 processing but on uncovering the principles of brain plasticity, the variables that modulate neural function, and the broader cognitive consequences of bilingualism. The theoretical contributions of these seminal works are profound, as they have provided empirical evidence to generate new theoretical frameworks and challenge existing paradigms in the field. Studies such as Morgan-Short et al. [3] and

Newman et al. [2] have demonstrated that proficiency and type of learning experience are critical factors that shape the L2 neurocognitive signature, moving the field beyond simple dichotomous comparisons toward a more nuanced understanding of experience-dependent plasticity. The work of [31, 55] on speech perception has contributed to developing new models of L2 phonological processing, while research by Hsu et al. [52] and Garbin et al. [54] has advanced our understanding of bilingual semantic organization. The theoretical review by Beatty-Martínez & Dussias [53] has helped reframe interpretations of core psycholinguistic phenomena, and the methodological innovations of Etard & Reichenbach [51] have opened new avenues for investigating neural dynamics in language processing. Furthermore, these studies have consistently emphasized the dynamic nature of neurocognitive processes in second language acquisition, demonstrating how neural signatures evolve with increasing proficiency and different learning experiences. The methodological rigor and theoretical significance of these works have established them as essential reading and mandatory references for subsequent research, solidifying their foundational status while driving the field toward new investigative directions and conceptual frameworks.

4.5.3 Key themes and findings

An in-depth analysis of the most influential papers reveals several interconnected key themes that chart the evolution of the field's understanding. A primary theme concerns the fundamental organization of the bilingual brain, with seminal works providing crucial neurophysiological evidence for how L2 processing can differ qualitatively from L1. Studies on speech perception by [31, 55] found reduced Mismatch Negativity (MMN) in L2 learners, indicating less automatic pre-attentive discrimination of L2 sounds. In the domain of syntax and semantics, Morgan-Short et al. [3] demonstrated that implicit versus explicit training leads to distinct ERP signatures (N400 vs. P600), showing that learning mechanism shapes neurocognitive processes. Newman et al. [2] further showed that proficiency modulates syntactic processing, with higher proficiency leading to a more native-like P600 response.

This research refutes a simplistic L1/L2 binary, establishing proficiency as a critical continuous variable that dynamically shapes neural signatures. The longitudinal work of Etard & Reichenbach [51] is particularly significant, tracking the same learners to document a neurocognitive shift from an N400 (semantic) to a P600 (syntactic) response to grammatical violations as proficiency increased. This provides powerful evidence for experience-dependent neuroplasticity and functional convergence toward L1-like processing. A core theme explores how L1 and L2 words are represented and connected in the brain. Studies such as those by Hsu et al. [52] and Garbin et al. [54] used the N400 component in cross-language priming paradigms to demonstrate a deeply integrated lexico-semantic system. Their findings that the N400 priming effect is modulated by semantic similarity provide crucial neural evidence for non-selective access models, where words from both languages actively compete within a shared neural space. Another highly influential theme explores the consequences of bilingualism beyond language itself. The study by Conrad et al. [50] investigated this by examining musicianship in bilinguals, finding that musically trained bilinguals showed enhanced N2 amplitudes (associated with conflict monitoring) during executive control tasks. This provides neural evidence that bilingualism, especially when combined with other cognitive training,

can enhance domain-general executive functions, thereby contributing to the ongoing debate on bilingual advantage.

Finally, a meta-theme across these papers is how methodological advancement drives theoretical progress. The field's evolution is marked by key innovations represented in the top-cited works. Etard & Reichenbach [51] introduced time–frequency analysis to link L2 learning to gamma-band oscillations, offering a dynamic view beyond ERPs. Furthermore, the highly influential theoretical review by Beatty-Martínez & Dussias [53] on the functional significance of the N400 reshaped interpretations of this component, demonstrating how refining methodological understanding is fundamental to theoretical progress across cognitive science.

4.5.4 Research implications and temporal evolution

The temporal distribution of these top-cited works shows clear intellectual progression. The early foundational studies (2010–2012), such as those by Wu & Thierry [31], Morgan-Short et al. [3], and Van Hell & Tokowicz [7], focused on establishing core ERP effects (MMN, N400, P600) and mapping basic differences between L1 and L2 processing. This was followed by a period of consolidation and exploration of the bilingual lexicon, e.g., Garbin et al. [54], Hsu et al. [52]. The most recent highly influential papers in this cohort (2017–2019) reflect the field's maturation, exploring more complex themes such as the theoretical interpretation of components [53], the dynamics of learning through longitudinal designs [51], and the analysis of neural oscillations. This evolution reflects a shift from mapping basic neurocognitive differences to understanding the dynamic, multifactorial, and experience-dependent nature of the neural architecture that supports second language acquisition and use.

4.6 International research collaboration patterns

The analysis of international research collaboration within the EEG-L2 processing research domain reveals a sophisticated and multi-layered network of scholarly partnerships that has been fundamentally instrumental in advancing the neurocognitive study of second language acquisition. Utilizing comprehensive bibliometric mapping techniques with a minimum collaboration threshold of three documents per country, the analysis identified 24 nations that have made substantial contributions to this specialized field. The resulting network visualization, presented in Fig. 6, illustrates the complex architecture of research relationships that has evolved over the past fifteen years, demonstrating the increasingly globalized nature of contemporary cognitive neuroscience research.

The network analysis reveals four principal collaborative clusters, each forming distinct strategic alliances shaped by shared research goals, historical links, and funding structures. These clusters reflect a globalized research environment where both regional and trans-oceanic partnerships thrive. The first cluster demonstrates remarkable geographical diversity, comprising nations from South America (Argentina, Chile, Colombia), Europe (Austria, Italy, Switzerland), Asia (Hong Kong), and Oceania (Australia). Such a configuration suggests a paradigm shift beyond regional cooperation towards a model of globalized, strategic knowledge exchange. This partnership likely coalesces around highly specialized research niches or methodological approaches within EEG-L2 research that transcend continental boundaries. For instance, the collaboration between Argentina and Australia indicates a deliberate effort to pool expertise and resources

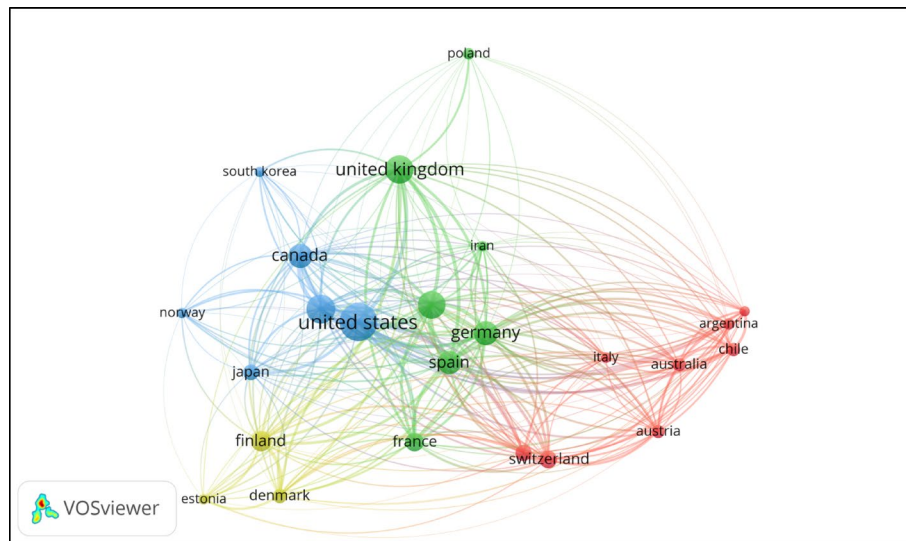


Fig. 6 International Collaboration Network Visualization. Nodes representing countries, with node size proportional to publication volume and line thickness indicating collaboration strength. Spatial proximity shows relational closeness

despite significant geographical distance, potentially driven by complementary strengths in specific neurocognitive theories or data analytics techniques.

The second cluster represents a dense, central network of European research powerhouses, including France, Germany, the Netherlands, Poland, Spain, and the United Kingdom, notably augmented by its connection to Iran. Germany and the United Kingdom undoubtedly function as primary hubs, a role supported by their extensive research infrastructure and high publication output. This cluster likely benefits profoundly from European Union framework programs that explicitly incentivize multinational research consortia. The inclusion of Iran demonstrates the cluster's outward reach and the role of scientific collaboration in building bridges across geopolitical divides, suggesting strong, sustained partnerships possibly focusing on shared linguistic interests or unique patient populations.

The third cluster is characterized by the dominant presence of the United States, which acts as the central node, consistent with its overarching influence in cognitive neuroscience and higher education. This cluster integrates key North American and Asia-Pacific research economies, highlighting the intense focus and investment in cognitive and neuroscientific research within these nations. The robust US-China collaboration is particularly salient, embodying the dual forces of scientific cooperation and strategic competition that define modern global research. The inclusion of Norway suggests a strong bilateral link between specific Norwegian research groups and this North America-Asia axis, possibly reflecting targeted expertise or unique data resources within the Norwegian research community.

In contrast to the expansive nature of the other clusters, the fourth group presents a compact, regionally coherent partnership consisting of Denmark, Estonia, and Finland. This tight-knit structure benefits from the Nordic-Baltic region's long-standing traditions of educational and scientific cooperation, facilitated by cultural and linguistic affinities, shared funding schemes, and geographically proximate institutions. The cluster's focus is likely highly specialized, leveraging shared expertise in areas such as language

acquisition pedagogy, educational technology, and cognitive processing, with particular emphasis on applying EEG methodologies in these domains. Its efficiency is derived from deep mutual understanding, streamlined communication, and complementary research specializations, allowing for highly integrated and productive collaboration.

The temporal evolution of these collaborations, depicted through color gradients in Fig. 7, reveals a fascinating chronological progression from 2016 to 2020. Earlier collaborations (2016–2017), represented by darker shades, were predominantly concentrated among Western nations with established neuroscience traditions. More recent collaborations (2019–2020), shown in lighter shades, indicate significantly increased engagement with Asian countries, particularly China, Japan, and South Korea. This temporal pattern reflects both the globalization of research excellence and strategic efforts to include more diverse linguistic populations in study samples, enhancing the ecological validity and generalizability of research findings.

The collaboration patterns reveal that while strong bilateral relationships continue to dominate the current research landscape, there is substantial potential for developing more complex multilateral partnerships. The high citation impact of research produced through collaborations between Western and Asian institutions suggests that combining diverse methodological expertise with access to different linguistic populations produces particularly impactful and innovative science. This temporal progression indicates a gradual but consistent shift toward more diverse and equitable international partnerships, reflecting both strategic institutional initiatives and the natural evolution of global research networks in cognitive neuroscience. The structural analysis further reveals that countries with high betweenness centrality particularly the United States (106.278), United Kingdom (60.429), and Spain (58.810) function as crucial knowledge brokers within the global network. These nations facilitate the flow of ideas and methodologies between otherwise disconnected research communities, accelerating innovation and cross-fertilization of concepts across geographical and disciplinary boundaries. This evolving collaboration landscape suggests that the future of EEG-L2 processing research will increasingly depend on the ability to forge and maintain diverse international

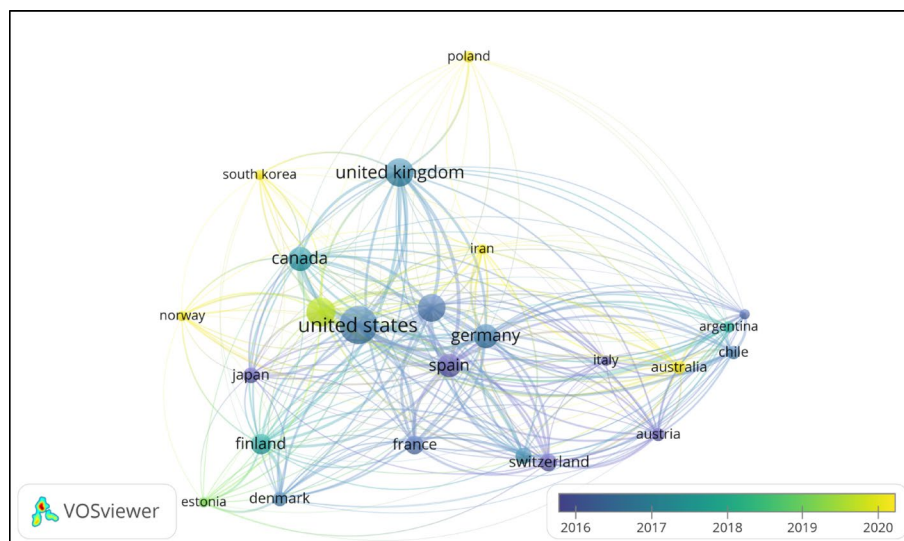


Fig. 7 Temporal Evolution of International Research Collaborations. Color gradients represent chronological progression from dark (earlier) to light (recent) collaborations

partnerships that leverage complementary strengths across different research traditions, methodological approaches, and linguistic backgrounds.

4.7 Keyword co-occurrence network analysis

The conceptual structure of the EEG-L2 processing research domain was examined through a comprehensive keyword co-occurrence analysis. The analysis began with examination of three keyword categories in the dataset: All Keywords (1996 terms), Author Keywords (1163 terms), and Index Keywords (833 terms). Author Keywords were selected for the co-occurrence network analysis due to their direct representation of authors' conceptualization of their research focus and more precise subject categorization compared to automated indexing terms.

Using VOSviewer's clustering algorithm, a minimum occurrence threshold of 5 was applied to the author keywords, resulting in 131 keywords meeting this criterion. This threshold optimization ensured both comprehensive coverage of significant research themes and clear visualization of conceptual relationships. The resulting network structure reveals 5836 links with a cumulative link strength of 36,652, indicating an extremely dense and deeply integrated research domain with fundamental and pervasive conceptual connections.

The comprehensive examination of keyword frequency patterns across the EEG-L2 processing research domain revealed a robust network of conceptual relationships. The analysis of author keywords, which more directly represent researchers' conceptual focus, provided a nuanced understanding of the field's intellectual structure. The application of a minimum occurrence threshold allowed for a focus on the most significant research themes, demonstrating the field's remarkable conceptual density and strong interdisciplinary connections.

4.7.1 Network structure and visualization analysis

The network structure, visualized in Fig. 8, reveals 5836 links with a cumulative link strength of 36,652, indicating an extremely dense and deeply integrated research domain with fundamental conceptual connections. The temporal evolution of research themes,

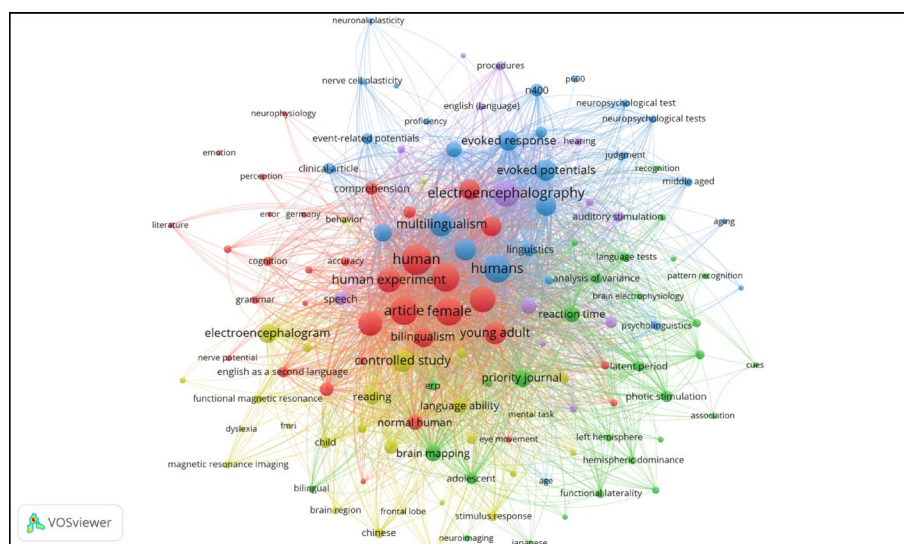


Fig. 8 Cluster-based network visualization of keyword co-occurrence

and specific linguistic mechanisms (Cluster 2), to refining methodological frameworks (Cluster 3), exploring broader cognitive processing interactions (Cluster 4), and accounting for key participant demographics (Cluster 5).

Complementing the network analysis, Fig. 10 presents a word cloud visualization generated using Biblioshiny, providing an alternative perspective on keyword prominence in EEG-L2 processing research. This visual representation emphasizes the central role of terms such as "electroencephalography," "language," and "bilingualism," while also drawing attention to emerging concepts within the field. The word cloud effectively highlights the most frequently occurring keywords, reinforcing key thematic areas identified in the network analysis and offering a clear, intuitive overview of the field's intellectual structure.

4.8 Temporal analysis of research topics

The temporal evolution of research topics in the EEG-L2 processing domain, as depicted in Fig. 11, reveals significant thematic shifts alongside a stable methodological core from 2010 to 2024. The analysis identifies distinct phases of development, beginning with the establishment of foundational methodological and demographic concepts. This Formative Phase (2010–2016) was characterized by the peak publication of studies establishing rigorous protocols, with key terms including "electroencephalography" (2016), "controlled study" (2014), "adult" (2016), "female" (2015), and "male" (2016). This period solidified a standardized research paradigm focused on employing EEG/ERP methodologies to map fundamental neurocognitive differences between first (L1) and second (L2) language processing, primarily in controlled laboratory settings.

The subsequent Expansion and Integration Phase (2017–2021) witnessed the maturation of this core paradigm and a significant broadening of theoretical scope. The rising prominence of terms like "n400" (2018), "semantics" (2018), and "language processing" (2017) reflected a deepened investigation into the neurophysiological substrates of specific linguistic phenomena. Concurrently, the integration of terms such as "executive function" (2020), "attention" (2020), and "bilingualism" (2019) marked a pivotal shift from purely linguistic comparisons to exploring the critical interplay between language control and domain-general cognitive systems. This phase established that L2 processing is not an isolated function but is deeply embedded within broader brain networks.

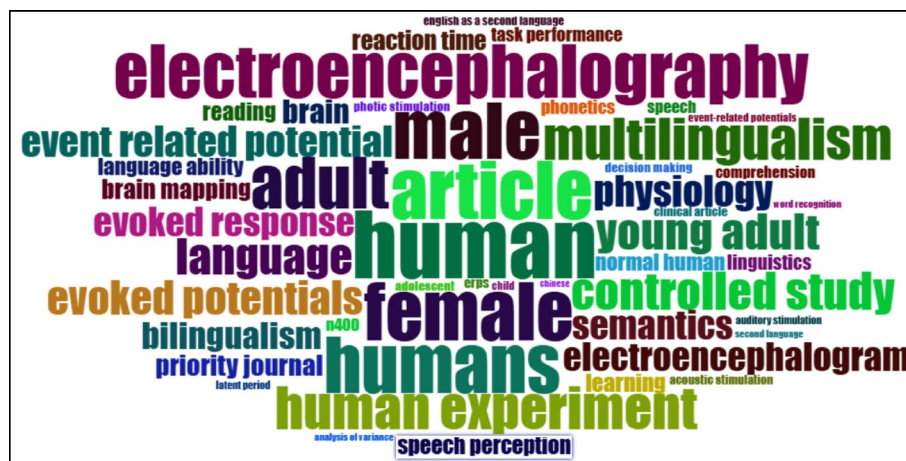


Fig. 10 Word cloud visualization of author keywords

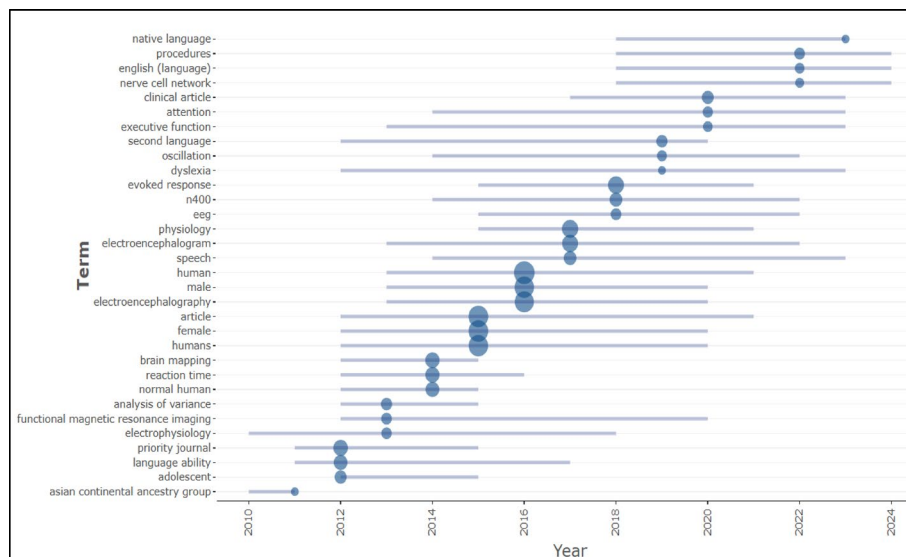


Fig. 11 Trending Research Topics

The current Sophistication and Diversification Phase (2022–2024) is defined by methodological innovation and theoretical nuance. The emergence of terms such as "oscillation" (2019), "eeg" (2018), and "nerve cell network" (2022) signifies a move beyond traditional ERP components, embracing time–frequency analysis and network-based approaches to understand the dynamic neural communication underlying language learning. Furthermore, the focus on terms like "native language" (2023), "procedures" (2022), and "english (language)" (2022) indicates a trend towards more precise linguistic comparisons and a heightened emphasis on methodological transparency and reproducibility, pointing toward more nuanced and specific research questions.

Overall, this temporal analysis underscores the dynamic yet consistent evolution of EEG-L2 processing research, shaped by methodological advancements and deeper theoretical inquiries. The field has successfully built upon a robust foundation to incorporate increasingly sophisticated theories and methods, reflecting a deeper investigation into the neurocognitive architecture of second language acquisition. Future research should continue to leverage these evolving methodologies to explore underexplored areas such as individual differences in neuroplasticity, the impact of learning contexts, and the neural correlations of language proficiency in diverse populations.

5 Discussion

This bibliometric analysis provides unprecedented insight into the intellectual architecture and evolutionary trajectory of EEG research in second language processing, revealing a field that has achieved remarkable theoretical sophistication and methodological diversity over the past fifteen years. The documented 4.29% annual growth rate, culminating in a peak of 18 publications in a select group of high-impact journals in 2024 [8, 9, 44, 56], demonstrates sustained scholarly momentum that transcends typical research cycles, indicating fundamental integration of neuroscientific approaches into mainstream second language acquisition research.

The emergence of distinct thematic clusters—neurocognitive foundations, linguistic mechanisms, methodological frameworks, cognitive processing, and demographic

factors—reflects the field's maturation from exploratory investigations to systematic theoretical frameworks. This clustering pattern suggests successful differentiation of research specializations while maintaining conceptual coherence, a hallmark of mature scientific disciplines. The dense keyword co-occurrence network (5836 links, 36,652 link strength) indicates robust theoretical integration, suggesting that disparate research threads are converging toward unified conceptual models of the bilingual brain.

The temporal progression from foundational ERP studies (2010–2016) [11, 57] through theoretical expansion (2017–2021) [58, 59] to current methodological sophistication (2022–2024) [9, 56] reveals a field undergoing systematic paradigmatic evolution. Early research established basic neurophysiological differences between L1 and L2 processing, particularly through canonical ERP components (N400, P600, LAN) [11, 57]. The subsequent integration of executive function and attention research (2017–2021) marked a crucial theoretical shift, recognizing that L2 processing is embedded within broader cognitive control networks rather than constituting an isolated linguistic module [14, 59]. The current emphasis on oscillatory dynamics and neural network approaches (2022–2024) represents a fundamental methodological paradigm shift from static, component-based analysis toward dynamic, systems-level understanding [9, 56]. This evolution aligns with broader trends in cognitive neuroscience toward network-based conceptualizations of brain function, suggesting that EEG-L2 research is successfully integrating cutting-edge neuroscientific methodologies.

The high citation impact of seminal works (seven publications exceeding 120 citations) demonstrates the field's capacity to produce foundational knowledge that shapes broader theoretical frameworks. The dominance of proficiency-related findings in top-cited papers suggests that experience-dependent neuroplasticity, rather than critical period constraints, represents the field's most influential theoretical contribution to second language acquisition theory [11, 60]. The international collaboration analysis reveals a sophisticated global research ecosystem characterized by strategic partnerships that transcend geographical constraints. The identification of four major collaborative clusters, spanning Europe, North America, Asia–Pacific, and South America, demonstrates successful globalization of research expertise and resources. The United States' central position (betweenness centrality: 106.278) as a knowledge broker, alongside the United Kingdom (60.429) and Germany (58.810), reflects both historical research leadership and contemporary investment in cognitive neuroscience infrastructure. The temporal shift from Western-dominated collaborations (2016–2017) toward increased Asian engagement (2019–2020) represents a strategic rebalancing of global research capacity. This evolution is theoretically significant because it enhances linguistic diversity in research samples, addressing longstanding concerns about the field's overreliance on Indo-European language pairs [45, 61]. The emergence of strong China–US collaborations, despite geopolitical tensions, underscores the transcendent nature of scientific cooperation in specialized domains [8, 44, 62]. The institutional leadership distribution—with Dutch, Swiss, and Argentine institutions dominating publication output—challenges traditional assumptions about research concentration in major anglophone research centers. This geographic diversification enhances theoretical robustness of the field by incorporating diverse linguistic typologies and educational contexts into empirical investigations [59, 63].

The methodological progression documented in this analysis reflects the field's successful adaptation to advancing neurotechnological capabilities. The evolution from basic ERP paradigms toward multivariate pattern analysis [30], machine learning approaches [8, 9, 44], and oscillatory dynamics [56, 64] represent more than technical advancement, it constitutes a profound theoretical shift in the conceptualization of how neural mechanisms support language learning. The integration of time–frequency analysis and neural oscillations research (evident in recent high-impact publications) enables investigation of dynamic neural communication patterns underlying language acquisition [4, 56]. This methodological sophistication addresses long-standing theoretical questions about the temporal coordination of neural networks during L2 processing, moving beyond static comparisons toward dynamic systems approaches. The consistent emphasis on controlled experimental designs (75 keyword occurrences) alongside emerging naturalistic approaches indicates methodological pluralism that balances ecological validity with experimental rigor. This balanced approach suggests the field has successfully navigated tensions between laboratory control and real-world relevance that have challenged other areas of cognitive research [21, 65].

The journal analysis reveals a stratified publication landscape where venue prestige strongly predicts research impact. The *Journal of Cognitive Neuroscience's* exceptional performance (13 articles, 874 citations) [3, 66] relative to more prolific outlets like *Neuropsychologia* (21 articles, 541 citations) [26, 67, 68] demonstrates the multiplicative effects of editorial selectivity on research dissemination. This pattern suggests that strategic publication significantly amplifies research impact within the cognitive neuroscience community. The concentration of high-impact research within elite venues (*Journal of Cognitive Neuroscience*, *Journal of Neuroscience*, *Cortex*) indicates that methodological rigor and theoretical significance serve as primary gatekeeping mechanisms. While this selective process ensures a high baseline of quality, it may also create barriers for emerging research themes that have not yet achieved methodological consensus, potentially slowing the dissemination of innovative but preliminary findings.

The bibliometric patterns suggest several emergent research frontiers that warrant strategic investment. The increasing prominence of individual differences research, reflected in demographic keyword clustering, indicates growing recognition that neuroplasticity varies systematically across learners [51, 60]. This personalized neurocognition approach could revolutionize L2 instruction through precision education interventions. The temporal emergence of network-based keywords suggests that future research will increasingly adopt systems neuroscience approaches, investigating how distributed brain networks coordinate during language learning [9, 56]. This shift toward connectivity analysis could elucidate mechanisms of cross-linguistic transfer and cognitive control that remain poorly understood. The growth in international collaborations, particularly those bridging linguistic typologies, positions the field to address universal versus language-specific aspects of L2 neurocognition [61, 69]. Such cross-linguistic investigations are essential for developing comprehensive theories of bilingual brain organization that transcend language pairs.

5.1 A bibliometrically-informed research agenda

Building directly upon the quantitative trends revealed by the present bibliometric analysis, several strategic future research directions emerge. **First**, the increasing

prominence of oscillatory and network-based keywords indicates a clear shift from static ERP contrasts toward dynamic systems and functional connectivity approaches. Future EEG studies are therefore expected to increasingly investigate how distributed neural networks interact during real-time L2 learning and processing.

Second, the growing clustering of demographic and learner-related keywords highlights the expanding importance of individual differences research. This trend points toward a future emphasis on modeling how variability in cognitive profiles, proficiency development, and learning environments shapes neural plasticity in second language acquisition. **Third**, the evolving structure of international collaboration networks suggests strong potential for developing more complex, multilateral, and cross-continental research partnerships that leverage complementary methodological, linguistic, and technological strengths. Such collaborations are especially critical for addressing understudied language pairs and enhancing cross-linguistic generalizability. **Finally**, the longitudinal evolution of thematic foci highlights the relative scarcity of long-term developmental investigations, identifying longitudinal neuroplasticity studies, learning-trajectory analyses, and research involving diverse linguistic populations as priority areas for future inquiry. Together, these bibliometrically grounded directions provide a coherent roadmap for the next phase of EEG-based second language research.

6 Limitations

Notwithstanding its robust contribution to mapping the intellectual structure of EEG-L2 processing research, this bibliometric analysis is subject to several methodological limitations that warrant acknowledgment. The reliance on Scopus as the sole database, despite its extensive coverage in cognitive neuroscience and applied linguistics, may have resulted in the omission of pertinent studies indexed in other major repositories such as Web of Science, PubMed, or ERIC, thereby potentially affecting the comprehensiveness and representativeness of the dataset. Additionally, while the search strategy was meticulously designed, its dependence on specific keywords within titles, abstracts, and author keywords might have overlooked relevant publications that employ synonymous terminology or where EEG methodology is not explicitly highlighted. The exclusion of non-journal sources, including books, book chapters, conference proceedings, and dissertations, may further introduce a source of bias, particularly toward overlooking emerging work and niche findings. Moreover, while bibliometric methods excel at identifying macro-level trends and impact, they are inherently quantitative and cannot capture the methodological rigor or theoretical subtleties of individual studies. Lastly, the use of citation metrics introduces a temporal bias, as recently published articles (e.g., from 2023 to 2024) may not yet have accrued sufficient citations to reflect their true scholarly influence. These limitations should be considered when interpreting the findings and generalizing the results.

7 Conclusion

This bibliometric analysis provides an overview of the evolution, trends, and impact of EEG-based second language processing research from 2010 to 2024. The study reveals a significant transformation in the field, driven by methodological innovations in neural oscillations analysis, theoretical advances in experience-dependent neuroplasticity, and the unprecedented expansion of international collaborative networks. The findings

highlight three distinct phases of research development: formative methodology establishment (2010–2016), expansion and integration of cognitive mechanisms (2017–2021), and sophistication through advanced analytical frameworks (2022–2024). These phases reflect the dynamic response of the global neurolinguistic community to the challenges and opportunities presented by the rapid advancement of EEG technologies and theoretical understanding of bilingual brain processing.

The analysis underscores the substantial increase in research output, with a robust annual growth rate of 4.29%, indicating the growing importance of EEG-L2 research as a critical area of cognitive neuroscience inquiry. The methodological evolution, in particular, has served as a catalyst, accelerating research activity and shifting focus toward sophisticated issues such as neural network dynamics, individual differences in neuroplasticity, and cross-linguistic transfer mechanisms. The contemporary period has cemented these trends, with research output maintaining a consistently high volume and increasingly sophisticated focus, suggesting a permanent shift in research priorities and the integration of advanced neuroscientific approaches into mainstream second language acquisition frameworks.

The study identifies key thematic clusters, including neurocognitive foundations, linguistic processing mechanisms, and methodological innovations, with a strong emphasis on executive control and experience-dependent adaptation. Radboud Universiteit, Universität Zürich, and Universidad Favaloro emerge as leading contributors to research output, highlighting the importance of specialized neurolinguistic expertise and advanced methodological knowledge in advancing EEG-L2 research. The analysis of most cited institutions reveals that strategic international partnerships and selective publication strategies achieved significant citation impacts, underscoring the importance of targeting high-impact neuroscience journals to enhance visibility and influence.

International collaboration, particularly spanning Europe, North America, Asia–Pacific, and South America, has been instrumental in advancing the field, with four major research clusters demonstrating successful transcendence of geographical boundaries. The temporal shift toward increased Asian engagement has enhanced linguistic diversity in research samples, addressing longstanding Indo-European language bias in bilingual studies, though opportunities for enhanced methodological standardization across research centers remain.

The citation analysis reveals a highly stratified distribution of research impact, with foundational theoretical frameworks achieving exceptional visibility and influence while recent methodological innovations await recognition. This disparity underscores the need for researchers to balance theoretical contribution with methodological rigor to enhance the cumulative impact of their work. The analysis also highlights the importance of addressing underexplored areas, such as longitudinal neuroplasticity studies, individual differences in learning trajectories, and diverse linguistic populations, to build a more robust and generalizable body of knowledge in bilingual neurocognition.

The findings of this study have several implications for researchers, funding agencies, and educational practitioners. First, continued investment in advanced EEG methodologies and cross-institutional collaboration is essential to address fundamental questions about neural mechanisms underlying multilingual competence, particularly in naturalistic learning contexts. Second, fostering international research partnerships, especially with institutions that have demonstrated high methodological sophistication, can

enhance the theoretical impact and practical applications of EEG-L2 research. Finally, the study calls for a more strategic approach to research planning, with emphasis on longitudinal designs and individual differences approaches, to ensure sustainable advancement in understanding experience-dependent neuroplasticity in language learning.

While this study provides valuable insights into the EEG-L2 research landscape, it is not without limitations. The reliance on Scopus data may exclude relevant research published in specialized neuroscience venues, and the focus on journal articles may overlook valuable methodological contributions in conference proceedings. Additionally, the temporal lag in citation accumulation may have introduced bias against recently published, potentially transformative research. Future bibliometric research could address these limitations by incorporating broader database coverage and exploring alternative impact metrics beyond traditional citation analysis.

In conclusion, this bibliometric analysis highlights the remarkable theoretical and methodological sophistication achieved in EEG-based second language processing research, particularly in response to advancing neurotechnological capabilities. By identifying key developmental phases, collaboration networks, and emerging research directions, this study contributes to the ongoing advancement of bilingual neurocognitive research and positions the field as a global leader in understanding multilingual brain function. The findings underscore the importance of continued methodological innovation, international collaboration, and theoretical integration to ultimately translate discoveries of the bilingual brain into effective, neuroscience-informed language learning frameworks and interventions for the twenty-first century.

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Author contributions

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